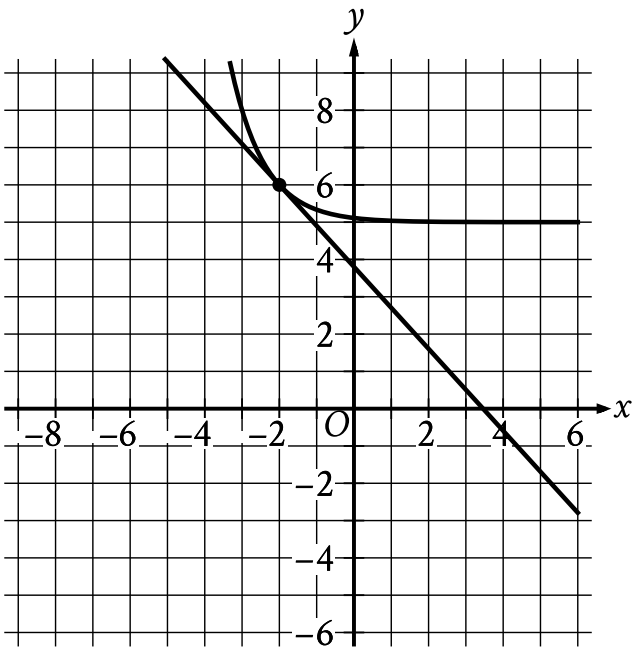


| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                        |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Easy <div><div></div><div></div><div></div></div> |

Question



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution  $(x, y)$  to this system?

Answer

- A.  $(6, 0)$
- B.  $(-2, 6)$
- C.  $(0, -2)$
- D.  $(0, 0)$

Correct Answer: B

Rationale

Choice B is correct. The solution  $(x, y)$  to the system of two equations corresponds to the point where the graphs of the equations intersect in the  $xy$ -plane. The graphs of the linear equation and the nonlinear equation shown intersect at the point  $(-2, 6)$ . Thus, the solution  $(x, y)$  to this system is  $(-2, 6)$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.